

Graph traversal

→ Note traversal of all vertices is not enough, you also need to traverse all the edges

Depth First Search

DFS(G, v)

begin

mark v ;

perform preWork on v ;

for all edges (v, w) do

if w is unmarked then

DFS(G, w)

perform postWork on (v, w) ;

end

Lemma: If G is connected, then all vertices will be marked and all its edges would be looked at atleast once

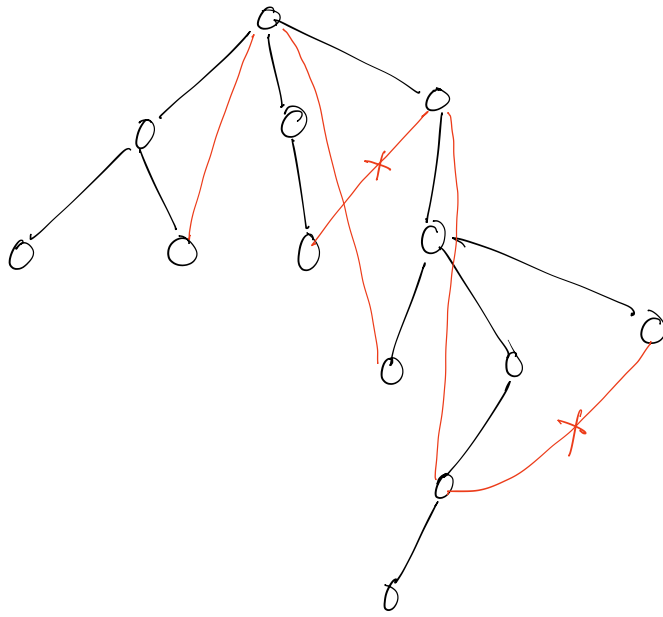
→ initially $DFS_NO := 1$;
use DFS with following pre Work

$$V.DFS = DFS_NO$$

$$DFS_NO = DFS_NO + 1;$$

→ use DFS with following post Work
if w was unmarked then add the edge (V, w) to T

Lemma: Let G be connected and undirected.
 T is a DFS tree of G formed by
 DFS. Every edge $e \in E$ either belongs to
 T or connects two vertices, one of
 which is an ancestor of the other.



we can't
 connect a child
 with a
 non-ancestor

— : DFS tree
 — : possible
 edges in
 original graph

If graph is connected
 then we will have a
 spanning tree, otherwise a
 spanning forest in general

Lemma: Let $G: (V, E)$ be a directed graph and let $T = (V, F)$ be a DFS tree of G . If $(v, w) \in E$ s.t. $v.\text{DFS_No} < w.\text{DFS_No}$ then w is a descendant of v in T

→ weakly connected: $U \rightarrow V$ connected if we remove direction
 → unilaterally connected: either $u \rightarrow v$ or $v \rightarrow u$
 → strongly connected: $u \rightarrow v$ and $v \rightarrow u$
 strongly connected $\Rightarrow$ unilaterally connected $\Rightarrow$ weakly connected

Time complexity = $O(V + E)$
 Space " = $O(V)$